

1 Appendix for Section 4

1.1 Growth Impatience Implies Harmenberg Impatience

We show here that **growth impatience** implies the condition imposed by ? to guarantee the existence of a permanent income weighted distribution of normalized market resources. Letting f denote the density of the permanent income shock ψ , the impatience condition imposed by ? is

$$\log(\mathbf{P}) < \int \log(\mathcal{G}\psi)\psi f(\psi) d\psi. \quad (1) \quad \{\text{eq:HarmImp}\}$$

Claim 1. *If growth impatience holds, then (1) holds.* {claim:gicimpliesh}

Proof. Since $\psi \mapsto \log(\mathcal{G}\psi)\psi$ is convex, Jensen's inequality implies

$$\int \log(\mathcal{G}\psi)\psi f(\psi) d\psi \geq \log(\mathcal{G}\mathbb{E}\psi)\mathbb{E}\psi = \log(\mathcal{G}) \quad (2) \quad \{\text{eq:Jensen}\}$$

Since **growth impatience** implies $\mathcal{G} > \mathbf{P}$ and $\log$ is strictly monotone increasing, the result follows. □

Remark 1. *The threshold in (1) is sharp: subtracting $\log \mathcal{G} = \log(\mathcal{G}\mathbb{E}[\psi])$ and using $\mathbb{E}[\psi] = 1$, (1) is equivalent to*

$$\log \mathbf{P}/\mathcal{G} < \mathbb{E}[\psi \log \psi] = D_{\text{KL}}(\psi f \parallel f) \geq 0,$$

*the Kullback–Leibler divergence between the permanent-income-weighted density ψf and f . Since this divergence is nonnegative (strictly positive whenever ψ is nondegenerate), **growth impatience** ($\log \mathbf{P}/\mathcal{G} < 0$) is sufficient but stronger than necessary; permanent-income risk relaxes the requirement.*

1.2 Apparent Balanced Growth in $\bar{c}$ and $\text{cov}(c, \mathbf{p})$

Section 4.2 demonstrates some propositions under the assumption that, when an economy satisfies the **GIC**, there will be constant growth factors $\Omega_{\bar{c}}$ and Ω_{cov} respectively for $\bar{c}$ (the average value of the consumption ratio) and $\text{cov}(c, \mathbf{p})$. In the case of a Szeidl-invariant economy, the main text shows that these are $\Omega_{\bar{c}} = 1$ and $\Omega_{\text{cov}} = \mathcal{G}$. If the economy is Harmenberg- but not Szeidl-invariant, no proof is offered that these growth factors will be constant. {sec:ApndxBalance}

1.3 $\log c$ and $\log \text{cov}(c, \mathbf{p})$ Grow Linearly

Figures 1 and 2 plot the results of simulations of an economy that satisfies Harmenberg- but not Szeidl-invariance with a population of 4 million agents over

the last 1000 periods (of a 2000 period simulation).¹ The first figure shows that $\log \bar{c}$ increases apparently linearly. The second figure shows that $\log(-\text{cov}(c, \mathbf{p}))$ also increases apparently linearly. (These results are produced by the notebook [ApndxBalancedGrowthcNrmAndCov.ipynb](#)).

¹For an exposition of our implementation of Harmenberg’s method, see [this supplemental appendix](#).

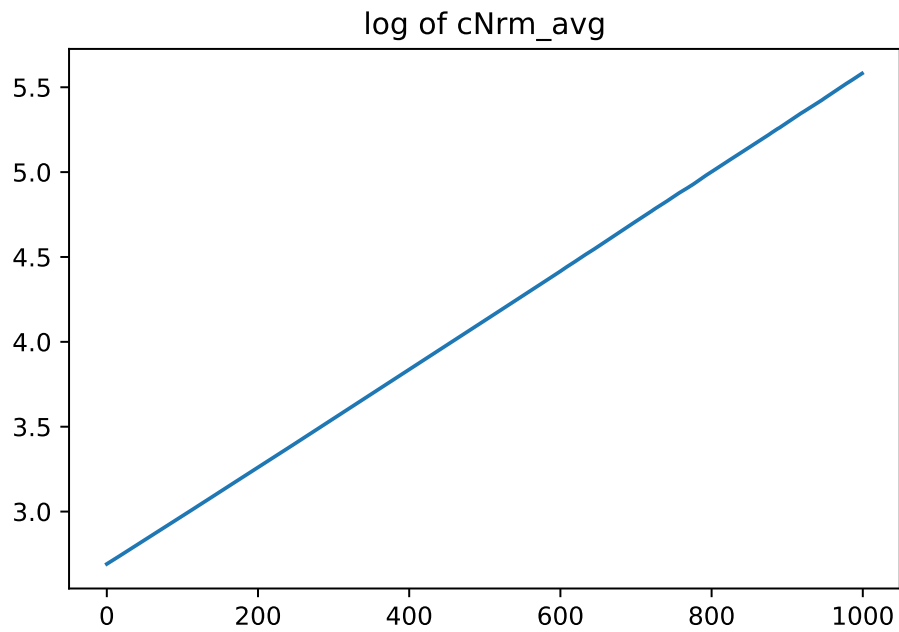


Figure 1 Appendix: $\log \mathfrak{c}$ Appears to Grow Linearly

{fig:logcNrm}

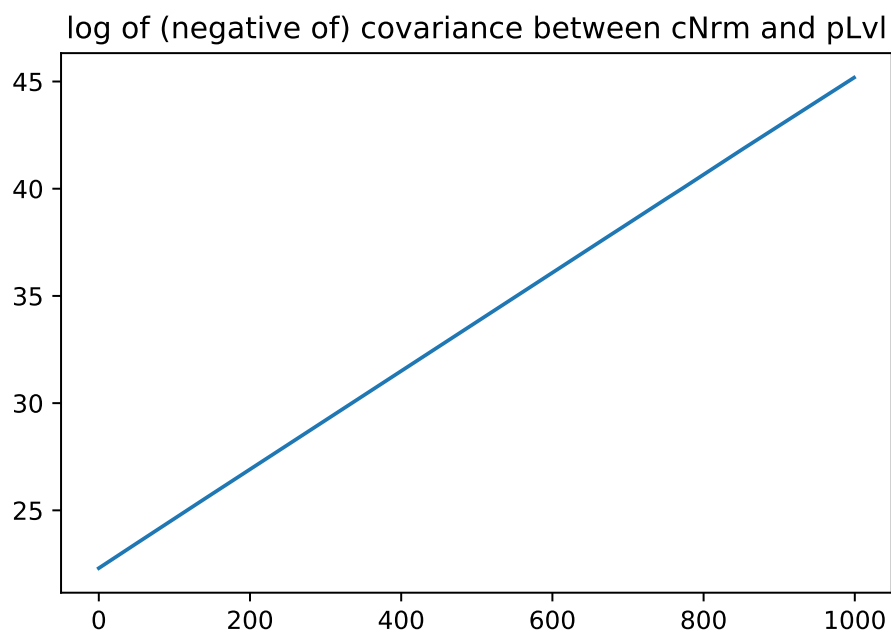


Figure 2 Appendix: $\log (-\text{cov}(c, \boldsymbol{p}))$ Appears to Grow Linearly

{fig:logcov}

(see [2](#))